

Course Code: 302-02
Course Title: Statistical Methods and Data Analysis

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Course Title	Statistical Methods and Data Analysis (Multi-Disciplinary Course – 03)
Credits	4
Course Category	Multidisciplinary Course (MDC-03)
Level of Course	200-299 (Intermediate Level Course)
Teaching per Week	4 Hrs.
Minimum weeks per Semester	15 (Including class work, examination, preparation etc.)
Review / Revision	-
Implementation Year:	A.Y. 2025-2026
Cognitive Skills of the Course	This course enhances critical thinking and problem-solving skills by encouraging students to analyze datasets, identify patterns, and apply statistical techniques in real-world scenarios. Students will develop analytical skills in selecting appropriate methods for data cleaning, transformation, and visualization. Additionally, they will learn to interpret statistical results and communicate findings effectively, fostering a deeper understanding of statistical concepts and data-driven decision-making. Remembering: The ability to recall facts, terms, and basic concepts from memory. Example: Listing statistical terms or recalling R functions. Understanding: The ability to explain ideas or concepts in your own words. Example: Describing the difference between mean and median. Application: Using learned concepts and knowledge in new or practical situations. Example: Applying data cleaning techniques in R to a new dataset. Analysis: Breaking down complex information into smaller parts and understanding relationships. Example: Analyzing a dataset to identify trends or outliers. Evaluation: Making judgments or forming opinions based on criteria and evidence. Assessing which statistical method is most appropriate for a specific dataset. Creation: Combining learned elements to form new patterns or structures. Example: Designing a new analysis pipeline or building a custom R function for data pre-processing.
Course Objective	1. To Develop a Strong Understanding of Statistical Concepts: Students will learn the fundamental concepts of statistics, including measures of central tendency, dispersion, and data sampling techniques, and how they are applied in data analysis. 2. To gain knowledge on Data Manipulation and Preprocessing: Students will gain proficiency in data cleaning, filtering, and transformation using R, preparing datasets for meaningful analysis. 3. To learn to Import, Visualize, and Analyze Data Using R: Students will acquire skills in importing data from various sources (CSV, SQLite), visualizing distributions, and performing statistical analysis through R. 4. To enhance Problem-Solving Skills through Data Visualization: Students will develop the ability to visualize and interpret univariate and bivariate data distributions using R's powerful plotting libraries like ggplot2. 5. To apply Statistical Analysis to Real-World Data: Students will be able to apply statistical methods to real-world datasets, conducting hypothesis testing, generating insights, and making data-driven decisions using R.
Pre-requisite	Knowledge of Fundamentals of Statistics and Mathematics of 10 th Grade Level

Course Outcomes	CO1: Understand foundational statistical concepts including descriptive statistics, probability theory, and basic inferential statistics. CO2: Apply statistical techniques such as hypothesis testing, confidence intervals, and correlation analysis to analyze and interpret data accurately. CO3: Demonstrate proficiency in data visualization methods to effectively communicate statistical findings and insights. CO4: Utilize basic statistical software tools or programming languages like R or Python to perform data analysis and visualization tasks. CO5: Develop critical thinking skills to assess the validity and reliability of statistical analyses and draw appropriate conclusions from data. CO6: Apply statistical reasoning to real-world scenarios and make informed decisions based on data-driven insights.									
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)		PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8	
	CO1									
	CO2									
	CO3									
	CO4									
	CO5									
	CO6									
Course Outcome	Upon completing the course, students will be able to apply statistical techniques using R to clean, analyse, and visualize data. They will develop skills in data manipulation, statistical analysis, and visualization, enabling them to make data-driven decisions and solve real-world problems.									
Course Content	Unit-1: Basic Statistical Concepts and Data Sampling Techniques: 1.1 Population vs. Sample: 1.1.1 Understanding the difference between population and sample, significance, and applications in statistics. 1.2 Central Tendency: 1.2.1 Measures of central tendency (mean, median, mode) and their use in statistical analysis. 1.3 Measures of Dispersion: 1.3.1 Understanding concept of Data Dispersion, skewness, correlation. 1.3.2 range, variance, standard deviation, and their importance in summarizing data. 1.4 Data Sampling Techniques: 1.4.1 Introduction to sampling methods: random sampling, stratified sampling, and their applications in data analysis. Unit-2: Introduction to R and Basic Data Representation : 2.1 Overview of R: 2.1.1 Concepts of R, its syntax, and applications in data analysis and statistics. 2.2 R Syntax: 2.2.1 Introduction to R syntax, basic operations, and expressions. 2.3 Variables and Data Types in R: 2.3.1 Understanding different data types in R (numeric, character, logical, etc.), and working with variables. 2.4 Data Importing in R: 2.4.1 Fetching data from CSV files and SQLite databases using R. 2.5 Data Representation in R: Visualizing data using histograms, box plots, and scatter plots. Unit-3: Data Preprocessing and Cleaning in R 3.1 Data Importing Techniques: 3.1.1 Fetching data from CSV files 3.1.2 Using read.csv(), read.table(), dplyr and RSQLite 3.1.2 SQLite databases into R. 3.2 Data Preprocessing in R:									

	3.2.1 Overview of preprocessing steps: 3.2.1.1 Cleaning, filtering 3.2.1.2 Transforming data for analysis. 3.3 Handling Missing Data: 3.3.1 Identifying and dealing with missing data values in R. 3.4 Data Cleaning and Transformation: 3.4.1 Techniques for data cleaning (removing duplicates, correcting inconsistencies), 3.4.2 Transforming data (type conversion, recoding variables). Unit-4: Data Manipulation and Visualization in R 4.1 Data Frame Manipulation: 4.1.1 Reordering, reshaping 4.1.2 Merging data frames in R. 4.2 Data Summary Statistics: 4.2.1 Calculating and summarizing key statistics (mean, median, mode, standard deviation) in R. 4.3 Visualizing Data Distributions: 4.3.1 Exploring and visualizing data distributions with R 4.3.2 Generating a Bell curve using normal distribution. 4.4 Graphical Representation of Univariate and Bivariate: 4.4.1 Univariate analysis and plot using ggplot2: 4.4.1.1 Box plot, Histogram, Bar plot, Density Plot 4.4.2 Bivariate analysis and plot using ggplot2: 4.4.2.1 Scatter plot, Box Plots, Line plots.
Reference Books	1. "An Introduction to Statistical Learning: with Applications in R" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, Publisher: Springer, ISBN: 978-1461471370 2. "R for Data Science: Import, Tidy, Transform, Visualize, and Model Data" by Hadley Wickham and Garrett Grolemund, Publisher: O'Reilly Media, ISBN: 978-1491910399 3. "Discovering Statistics Using R" by Andy Field, Jeremy Miles, and Zoe Field Publisher: SAGE Publications Ltd, ISBN: 978-1446200469 4. "Practical Data Science with R" by Nina Zumel and John Mount Publisher: Manning Publications, ISBN: 978-1617291562 5. "Statistics: Unlocking the Power of Data" by Robin H. Lock, Patti Frazer Lock, Kari Lock Morgan, and Eric F. Lock, Publisher: Wiley, ISBN: 978-1119325572 6. "The Art of R Programming: A Tour of Statistical Software Design" by Norman Matloff, Publisher: No Starch Press, ISBN: 978-1593273842 7. "Introduction to Probability and Statistics Using R" by G. Jay Kerns, Publisher: RStudio, PBC, ISBN: 978-1886529450 8. "Business Analytics – The science of Data-Driven Decision Making" by U.Dinesh Kumar, Publisher: Wiley, ISBN: 978-81-265-6872-2
Teaching Methodology	Class Work, Discussion, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.